

Design and Implementation of IoT Based Accident Detection and Prevention

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Abstract: In the modern world, an increase in the usage of automobiles for commercial purposes has also increased the number of accidents occurring in commercial vehicles, which leads to the loss of life of the people involved in the accident. To minimize the death rates involved in an accident, the people who are met with the accident must claim medical assistance at the correct time. Our paper is concerned with two set-ups. One set-up is associated with the vehicle, where the use of a MEMS or gyroscopic sensor, a vibration sensor, and a gas sensor integrated with Arduino helps us detect the accident, and the location is detected by the GPS module and updated in the cloud by using the ESP8266 Wi-Fi module. If any accident is detected, the RF transmitter circuit sends the signal to the RF receiver. The other configuration is related to the Ambulance which consists of an RF receiver circuit integrated with the NodeMCU microcontroller. When the signal reaches the receiver, NodeMCU retrieves the information from the cloud and displays it on the LCD. Integration of a tracking system with a Radiofrequency transmitter and receiver helps build IoT services using embedded systems. The system of providing medical assistance to the people involved in the accident would help us reduce the death rates.

Keywords--- Accidents, Arduino UNO, Vibration sensor, MEMS sensor, Radiofrequency transmitter and receiver, NodeMCU, Internet of Things, Angle detection, vibration detection.

I. INTRODUCTION

Our paper is most relevant in the current scenario, where the death rates due to road accidents are the major concern in accidental deaths occurring all over the world. According to the statistics of the World Health Organization, almost 1.3 billion people die every year as a result of road accidents. A large group of people, ranging between 20 and 50 million, are suffering non-fatal injuries, with many incurring a disability as a result of their injury. In countries like India, with one of the largest populations in the world, the occurrence of accidents is higher, and road accidents play a major part in the number of accidents all over India.

According to the statistics of the ministry of road transport and highways, during the year 2020, the total recorded 3,66,138 road accidents caused the deaths of 1,31,714 people and injured 3,48,279 people. The lives of everyone are important because they are not just as individuals; they are associated with their families, and their families are associated with society. This paper primarily focuses on providing the medical assistance to

the people who are involved in the accident at the correct time. In India, there are lots of blind spots on the highway and in densely populated rural areas where the occurrence of the accident may not be known. During such a critical condition, providing the correct medical attention to the people is extremely important; if not, it may cost the life of the person involved in the accident. Many precautions are being taken to protect people's lives as technology advances.

With all these measures, the number of road accidents is increasing every day. Accidents are unavoidable in a fast-moving world, so providing the correct medical assistance to the person involved in the accident will be the correct measure to reduce the death rates. IoT services are helpful in storing and processing the data in the cloud, and the data can be easily retrieved from the cloud within a few seconds. By updating the location information in the cloud using IoT services, the cloud benefits society by providing accurate medical assistance to the people involved in the accident.

Our paper proposes the operation of the IoT (Internet of Things) with the GPS module and wi-fi module, which are used as tracking systems in the automobile industry. With these existing GPS modules and Wi-Fi modules, the integration of the vibration sensor and gyroscopic sensor for accident detection and the gas sensor for alcohol consumption with the Arduino microcontroller can provide the exact location of the accident, which is updated in the cloud. The implementation of an RF transmitter in Arduino helps send the signal to an RF receiver, which is in the ambulance. The NodeMCU associated with the RF receiver helps retrieve the information from the cloud. This helps provide the correct medical assistance to people involved in the accident at the correct time.

II. LITERATURE SURVEY AND ANALYSIS

In the modern era of transportation, everyone is working hard to reduce the number of accidents with the available technology. Certainly, some of the past works that improved safety measures and reduced accidents are tabulated below. It begins with GPS navigation, GSM communication, IoT technology, the use of different sensors, a machine learning model, a deep learning model, a neural network model, etc. In different ways, there are many processes being undergone to reduce the rate of accidents.

Our paper generally has the integration of the GPS navigation, which is used to detect the geographical

Comparative Analysis of Literature Works			
S.NO.	Papers	Year	Remarks
1.	Ubiquitous GPS vehicle tracking and management system.	2011	It provides a vehicle tracking system through the GPS, GPRS, and GSM communication networks.
2.	Improving Estimation of Vehicle's Trajectory Using the Latest Global Positioning System with Kalman Filtering.	2011	It proposes the Kalman filters and the "interacting multiple model" be used to provide the future location of the automobile.
3.	Accident Detection Using Convolutional Neural Networks.	2019	In this paper, CCTV cameras on highways are used through a deep learning convolution neural network model for accident detection.
4.	Accident Detection and Warning System.	2021	In this paper, CCTV cameras are enhanced with night vision that captures the video and produces the real-time signals that are used to detect the accident.
5.	A Road Accident Prediction Model Using Data Mining Techniques.	2021	An accident prediction model is created using the Apriori algorithm and vector machines as part of a data mining technique to prevent accidents.
6.	Predictive Analytics of Road Accidents Using Machine Learning.	2022	A machine-learning approach in terms of considering factors like the number of accidents in a year, month, state, etc. is considered and used to improve the safety measures.
7.	Prototype Proposal for Quick Accident Detection and Response System (QARS).	2020	In order to develop QARS with the help of the Internet of Things, we acquired the information using the GPS and GSM communication networks.
8.	Car Accident Prevention and Health Monitoring System for Drivers.	2021	In this proposal, the health monitoring sensors are used to monitor the health of the driver and the movement of the vehicle.

Fig.1. Literature survey comparison

location of the vehicle. If an accident is detected, the RF transmitter in the vehicle is activated and searches for the nearby receiver circuit and conveys the location information through the IoT. In the real-time implementation of the RF circuit, use the same air medium as in the radio station. So, the transmission will be accurate, and any errors can be eliminated.

III. COMPONENTS AND WORKING

A. Arduino

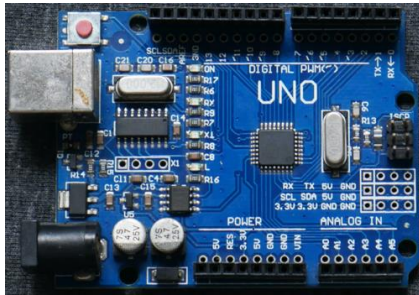


Fig.2. Arduino

Arduino UNO is a microcontroller based on a datasheet named ATmega328. Its operating voltage is 5V. The recommended input storage is 7-12V. The limits of input voltage are 6-20V. This microcontroller board consist of 14 digital I/O pins. The DC current per I/O pin is 40mA. It also contains flash memory, SRAM, EEPROM, and Clock speed. It is an open-source microcontroller that is flexible and easily programmable. Through this board, we may interface with various expansion boards and several other circuits. Arduino UNO boards are used in many applications, such as weighing machines, automation, embedded systems, etc. The Arduino microcontroller is the brain of the IoT, which receives and processes the information from the sensor. It is very easy to code in Arduino using the IDE platform and to perform different functions.

B. Global positioning sensor (GPS)



Fig.3. Global positioning sensor

GPS modules are the universally used navigation sensor that utilizes satellite communication in the orbit around the earth to provide the required information. GPS modules are used to receive data that is directly sent from satellites through RF frequencies. Receiving data, it contains tiny processors and antennas. Through these GPS modules, we could locate, navigate, and trace any objects or personal movements. In our system, we use a GPS module to locate the vehicle and send it to the receiver end.

C. Vibration sensor



Fig.4. Vibration sensor

A Vibration sensor is a device that is used to measure the vibration level in any object. It contains accelerometers that sense the vibration. The vibrations can be measured in any object through these vibration sensors. The ceramic piezoelectric sensor otherwise called an accelerometer is mostly used to measure acceleration. In the vibration sensor, we have transducers. These transducers convert the mechanical forces which are caused due to vibration into an electrical current by the piezoelectric effect. The range of sensitivity of these sensors is 10mV/g to 100mV/g. There are many vibration sensors like accelerometer sensors, strain gauge sensors, velocity sensors, gyroscope sensors, pressure or microphone sensor, laser displacement sensor, capacitive displacement, vibration meters, and vibration data loggers. Based on application, we can make use of these vibration sensors to sense and measure the vibration. Here, in our system, we use a vibration sensor that is directly related to accelerometers that are used to detect the accident.

D. Angle detection sensor



Fig.5. Angle detection sensor

An Angle detection sensor is a device that detects the change in the angle of a certain object and sends the data. MEMS technology is composed of sensors suspended between a pair of capacitive plates. These sensor measures and converts the mechanical rotation into an electrical signal. In our system, these signals are fed into the MEMS sensors (Micro-electrochemical systems) which is a tiny chip that combines mechanical and electrical signals. MEMS sensor are selected for the detection of change in the axes of the vehicle due to the following advantages they can withstand extreme temperature, shock resistance and, vibration resistance. In MEMS the angle of axes can be changed which makes it more flexible. Here we use an angle sensor to detect the angle of the vehicle. If any angle deviation occurs the sensor is activated which produces the alert signal.

E. Gas sensor

Gas sensors are the sensors that are used to detect hazardous gases and vapors. The presence and concentration of the gases can be detected using these gas sensors. There are many types of gas sensors. In these types, MQ-type gas sensor consist of 6 terminals where 4 terminals (A, A, B, B) are used as input or output and the other 2 terminals (H, H) are used for heating the coil. The gas sensor module consists of 4 terminals. They are Vcc, GND, Digital outputs, and Analog outputs. To detect the consumption of the alcohol gas sensor are used in the system.

F. ESP8266 Wi-Fi module

The name itself says that ESP8266 is a Wi-Fi module, which is a Wi-Fi microchip. It has built-in TCP/IP networking software. This Wi-Fi module is used to send and receive data over Wi-Fi. The Internet Of Things uses these modules to communicate between two devices. ESP8266 is the cheapest device that can be used in many systems. ESP8266 Wi-Fi module consists of 17 GPIO pins. These modules allow microcontrollers to connect to 2.4 GHz Wi-Fi.

G. RF Transmitter and Receiver

RF module is a Radio Frequency module that uses a Radio Frequency signal. The frequency of these RF transmitters and receiver ranges between 30kHz & 300kHz. RF module consists of both an RF transmitter and receiver. The transmitter is Tx and the receiver is Rx. The pair operates with a frequency of 433mHz. The voltage used in a transmitter is between 3V to 6V. The transmitter and receiver is a one-way communication

that is the transmitter can only send data and the receiver can only receive the transmitted data. To get more inputs and outputs in the RF transmitter module, we use an encoder and decoder.

So, through this RF transmitter and receiver we could send data through the air medium. Thus in our system, we integrate an angle sensor, vibration sensor, and gas sensor and if any deviations occurred, these sensors send the data to the receiver end.

H. LCD display

LCD, also known as Liquid Crystal display is a flat panel display that is used to display numbers and alphabets. It consumes less power as it works on the principle that it blocks the light rather than dissipates it. It uses the principle that when an electric current is passed to liquid crystal molecules, they untwist. This makes the light angle change. This causes a change in the angle of the top polarizing filter. Therefore, little light is allowed to pass only in a particular area in the LCD. Thus, the area becomes darker. The LCD contains several layers such as a mirror, polarizing film, glass filter, a negative electrode, liquid crystal layer, positive electrode, glass filter, and cover glass and the cover glass displays the image.

I. NodeMCU

NodeMCU is a low-level IoT platform where the basic levels of the software code are integrated into the hardware-specified device. It is one of the microcontroller unit integration with the node for retrieving the data from the cloud. NodeMCU is a microcontroller with an in-built Wi-Fi module, that acts in either way of controlling the system also retrieving the data from the cloud. In this system, NodeMCU is the microcontroller unit for the RF receiver and also, acts as the IoT platform that retrieves the data from the cloud and displays it in the ambulance for medical assistance.

IV. WORKING PRINCIPLE

The working of the setup is generally defined in different steps. The operational setup of the system is separated into two setups as the transmitter setup which is placed in the vehicle consists of a vibration sensor, gas sensor, MEMS sensor, ESP8266 Wi-Fi module, GPS module, and RF transmitter connected to Arduino microcontroller. The receiver setup which is placed in the ambulance consists of an LCD and RF receiver connected to a NodeMCU microcontroller.

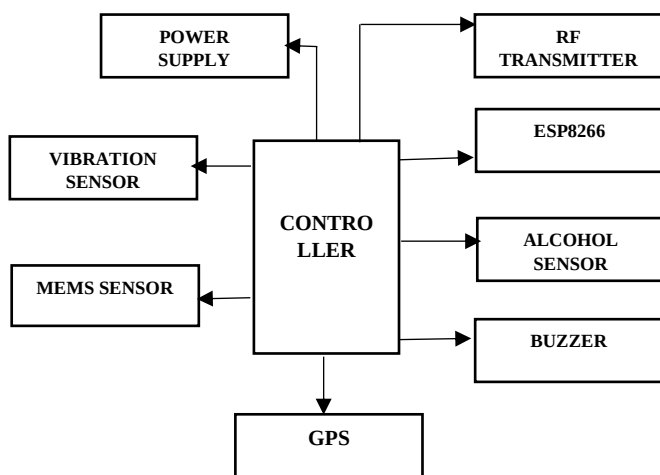


Fig.6. Vehicle setup

The sequential operation of this setup starts with the Arduino microcontroller which works on the principle of universal asynchronous receiver transmitter (UART) protocol. The first step is to build the sensor network where the sensors, GPS, Wi-Fi module, and RF transmitter are connected to the Arduino with an operating voltage of 5V, except the MEMS sensor which operates at a voltage of 3.3V. This sensor network is the conventional network where the information from the sensors and GPS are stored in the ESP8266 Wi-Fi module through the Arduino microcontroller. After building the sensor connections in the transmitter setup, the location is updated in the Wi-Fi module, which is then stored in the cloud through the IoT. IoT helps us to access a wide range of embedded devices and provides web services. Here, Thing Speak acts as the API for the IoT and the provided information is stored in the cloud. Thing speak is programmed in a way of establishing connections between the sensor network, Wi-Fi module, and cloud. Therefore, the program is fed to Arduino. With this establishment of a sensor network and IoT, if the accident is detected in the transmitter setup, the location is updated in the cloud and the buzzer connected to the Arduino starts to produce the alert signal and waits for the manual reset button. In case the alert signal from the buzzer is not resisted, then the RF transmitter connected to the vehicle starts to send the radio waves through the air medium with a certain frequency. RF signals are modulated with the carrier wave modulation in the transmitter to reduce the error or noise in the RF signal.

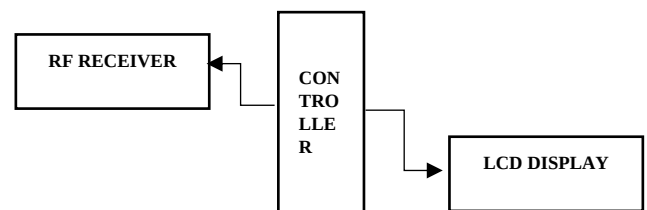


Fig.7. Ambulance Setup

The signal from the RF transmitter searches for the nearby RF receiver for receiving the signal. When the signal is received by the RF receiver of the same frequency in the ambulance setup. The RF receiver is connected to the NodeMCU microcontroller which is used to fetch the vehicle's geographical information from the cloud. The final step is to display the information on an LCD. The LCD that is connected to the NodeMCU, displays the latitude and the longitude of the vehicle in the ambulance. The prevention protocol of the setup is related to the Gas sensor that operates in the 5V which is used to detect the consumption of alcohol. If the driver is consumed alcohol the buzzer will produce the alert sound and the vehicle is immobile. Arduino and NodeMCU are microcontroller-based embedded system which is used to connect the physical device and store the information in the cloud using IoT services through the Thing Speak platform.

S.NO.	Comparison	Existing Technology	Revisionism
1.	GSM vs RF	GSM is used as the alert system for emergency contacts which doesn't provide immediate aid to the person involved in the accident.	RF transmitter and receiver is the alert system implemented in the ambulance in which immediate aid is provided to the person involved in the accident
2.	Machine learning vs RF	It needs a lot of data for study and interpretation which can cause some errors in the real-time application.	RF uses radio waves which are modulated and so the errors can be eliminated in real-time.
3.	Deep learning vs RF	Deep learning requires complex data models and expensive	It doesn't interfere with the other radio waves because each wave is categorized with

		GPUs for training.	different frequencies.
4.	Fuzzy logic vs RF	Fuzzy algorithms and control system are not self-reliable, it needs some human assistance.	The Logic for RF is generally concerned with the microcontroller connection and it is reliable.
5.	Data mining vs RF	Larger databases are required for the processing which is not concerned with privacy and security.	RF circuit doesn't need any server which can be processed through the antennas in the vehicle.

Fig.8. Comparative analysis for novelty

In this system, the use of the radio frequency transmitter and receiver is the unique approach to sending information to the ambulance that is used to provide immediate medical assistance to the person involved in the accident. RF transmitter and receiver are the better approaches for transmitting accident information because it is faster and more accurate and also it doesn't require larger data for processing as compared to other systems. In the real-time implementation of the RF transmitter and receiver, the signal is transmitted along with the carrier wave using a modulation technic. This, modulation technic makes the system more accurate and the signal is transmitted without any noise and interference.

V. Flow chart

The sequential order of operation of the system is described in the flowchart. The system of operation starts with the initialization of the sensors, GPS and Wi-Fi module, and RF transmitter and receiver. The next step is that if the system is normal, it does not proceed to the next step but if there is any kind of accident occurred then the buzzer is on which indicates an alert signal. The next step is that if the reset button is activated manually messaging is terminated. In case the buzzer is not activated then the next steps have proceeded. The following steps are geographical location are deducted using the GPS module 9 and updated in the cloud using the ESP8266 Wi-Fi module. The NodeMCU microcontroller retrieves the geographical information and displays it in the ambulance. The last step is that with that provided information the ambulance reaches the accident spot and provides medical assistance to the people involved in the accident. This is the step-by-step operation of the system.

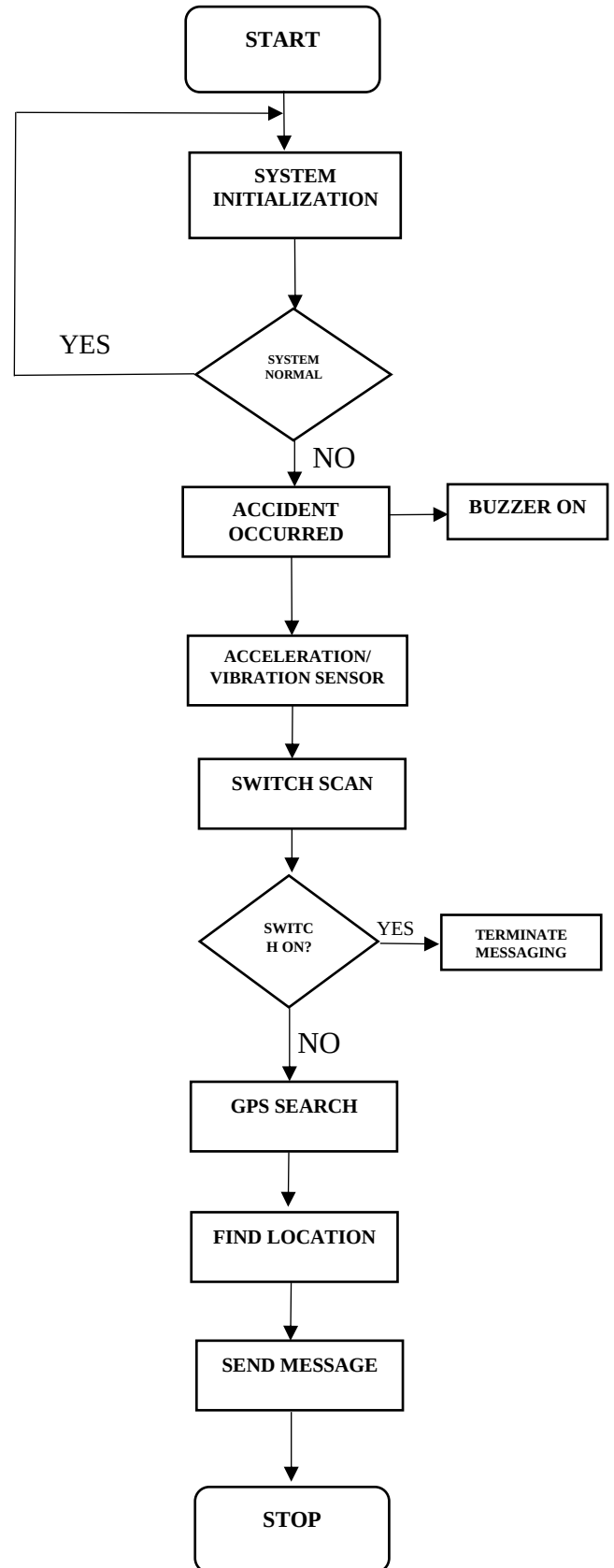


Fig.9. Flow chart

VI. HARDWARE SETUP

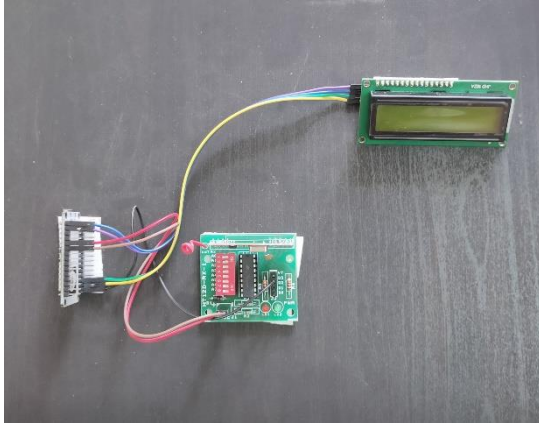


Fig.10. Receiver Setup

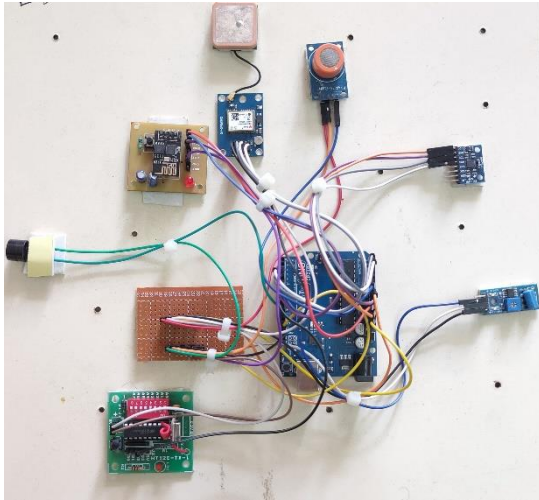


Fig.11. Transmitter Setup

VII. CONCLUSION

Road accidents are the major cause of the increasing death rates. To reduce the death rates, we are presenting an approach to detect the accident and provide immediate medical aid to the person involved in the accident which would considerably reduce the mortality rates and fatal injuries. The location is detected using a GPS module and uploaded in the cloud using the ESP8266 module and IoT services, accident is detected using a vibration sensor and angle detection sensor in either way and operates the system where the RF transmitter sends the radio waves and searches for the nearby receiver. The RF receiver receives the radio waves and operates the NodeMCU microcontroller which is used to fetch the information from the cloud and display it in the LCD placed in the ambulance. The use of Radio Frequency

communication uses the air medium for transmission which eliminates any other sources for transmission. The error in the radio waves is eliminated using the modulation technique in which the radio waves are amplified with a carrier wave of larger frequencies. Thus, the use of radio frequency transmission will be the efficient and effective way of transmitting the signal because compared to the earlier approaches it doesn't require larger data, complex models, control systems, servers, and databases. The system can be concluded as, in real-time implementation the use of an RF transmitter and receiver and the use of two sensors for accident detection has the unique approach to reducing the death rates in accidents.

Radio wave transmission is an important aspect of communication because of the longer wavelength and traveling speed. There are different technologies adopted in the automobile industry to make vehicles more intelligent. Therefore, the addition of an accident detection system in the automobile industry will reduce the death rates and improves safety measures. In the future, integration of radio wave transmission with different technologies like satellite communication, 5G network, and artificial intelligence will make the vehicle more intelligent and safe.

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